Project Design Document

Project title: Script-Controlled ACL to Restrict Record Access Based on Field Value in ServiceNow

Members	Name	NM ID
Team Leader	Divyadharshini A	Asaluau1324129008
Team Member 1	Karthiga M	Asaluau1324129016
Team Member 2	Samiha A	Asaluau1324129033
Team Member 3	Balasri R	Asaluau1324129003
Team Member 4	Divya P	Asaluau1324129007

ABSTRACT

The project design defines the structure of a ServiceNow-based access control system for protecting student/institution records. The system uses users, custom roles, ACLs, and a field-based condition to control access to records.

The design provides separate permissions for Read, Write, Create, and Delete operations. The Read operation is further restricted based on the **Branch = EEE** condition.

SYSTEM DESIGN

The system consists of the following major components:

- **User:** EEE User
- **Roles:** `bb1`, `bb2`, `bb3`, `bb4`
- **Table:** Institution/Student Details
- **ACLs:** Read, Write, Create, Delete
- **Field Condition:** Branch = EEE
- **Script:** Used in the Read ACL for role verification

Basic System Flow

User → Role Verification → ACL → Condition/Script → Allow or Deny Access

TABLE DESIGN

The system uses an Institution/Student Details table with the following fields:

Field Name	Purpose
Student Roll Number	Auto-generated student number
Student Name	Stores student name
Faculty Name	Stores faculty information
Branch	Stores branch information
Phone Number	Stores contact number
Description	Stores additional details

The **Branch** field is important because it is used for the Read access restriction.

ROLE DESIGN

Role	**Permission**
bb1	Read
bb2	Write
bb3	Create
bb4	Delete

ACL DESIGN

The ACL structure is:

ACL Operation Role Condition Script

Read ACL Read `bb1` Branch = EEE Yes

Write ACL Write `bb2` None No

Create ACL Create `bb4` None No

Delete ACL Delete `bb4` None No

The Read ACL uses a script to verify the user's role, while the other ACLs use role-based access without scripting.

SECURITY DESIGN

The system follows role-based and field-based access control.

- `bb1` controls Read access.
- `bb2` controls Write access.
- `bb3` controls Create
- bb4 Delete access.
- Read access is restricted using **Branch = EEE**.
- Users without the required role are denied the corresponding operation.

This design helps protect student information from unauthorized access or modification.

ACCESS CONTROL FLOW

User

↓

Check User Role

↓

Check ACL

↓

Check Branch Condition / Script

↓

Allow or Deny Access

For example:

EEE User → `bb1` → Branch = EEE → Read Access

EXPECTED OUTCOME

The designed system should provide controlled access to student records based on user roles and the required branch condition.

The final access structure is:

`bb1` → Read

`bb2` → Write

bb3 → Create

bb4 ---→ Delete

Branch = EEE → Read restriction

CONCLUSION

The project design provides a structured approach for implementing script-controlled ACLs in ServiceNow. The use of separate roles for Read, Write, Create, and Delete operations, along with the **Branch = EEE** condition, helps control access to student records. The design provides a simple and secure structure for preventing unauthorized access and protecting institutional information.